AI Patent Intelligence: Week 1 report

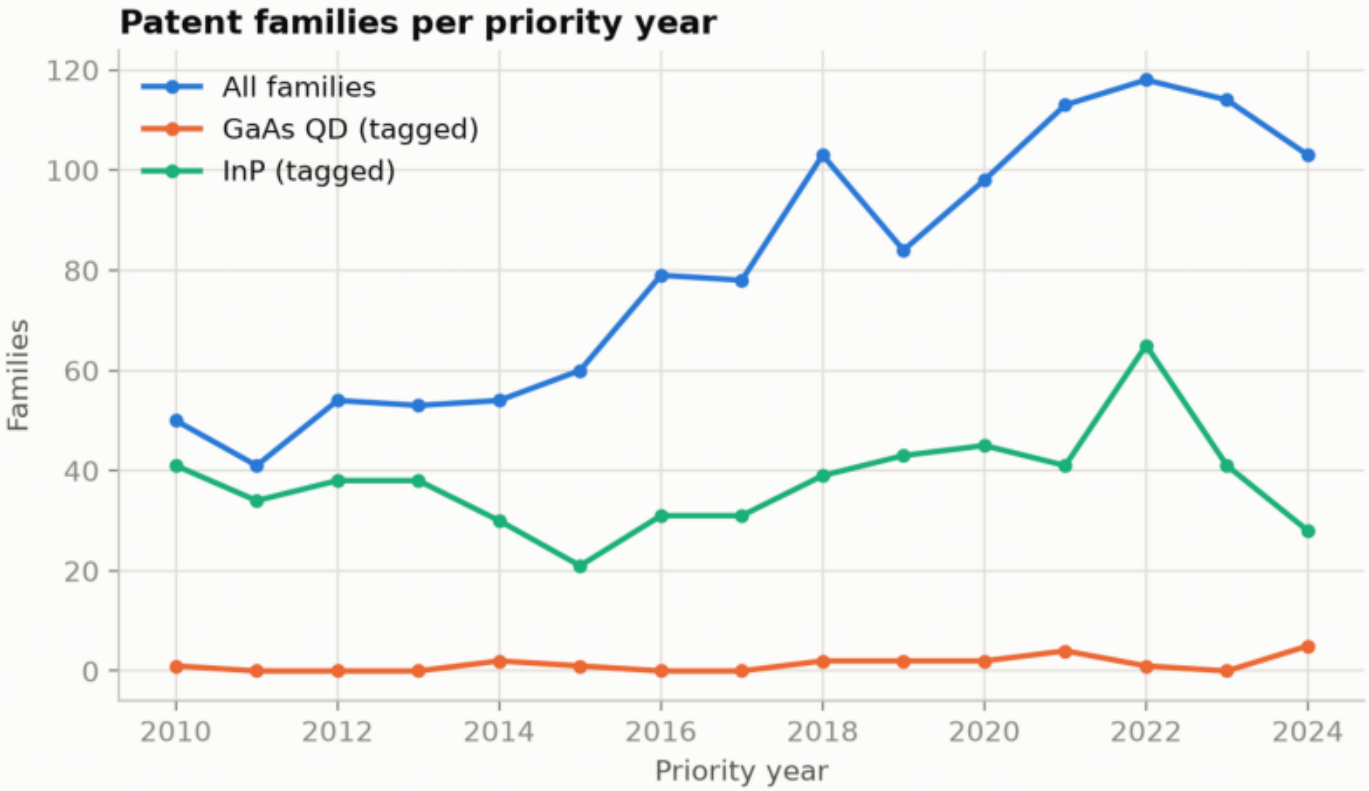
Dataset: 2248 patent documents from the EPO Open Patent Services (publication years 2010 to 2026), grouped into 1540 patent families.

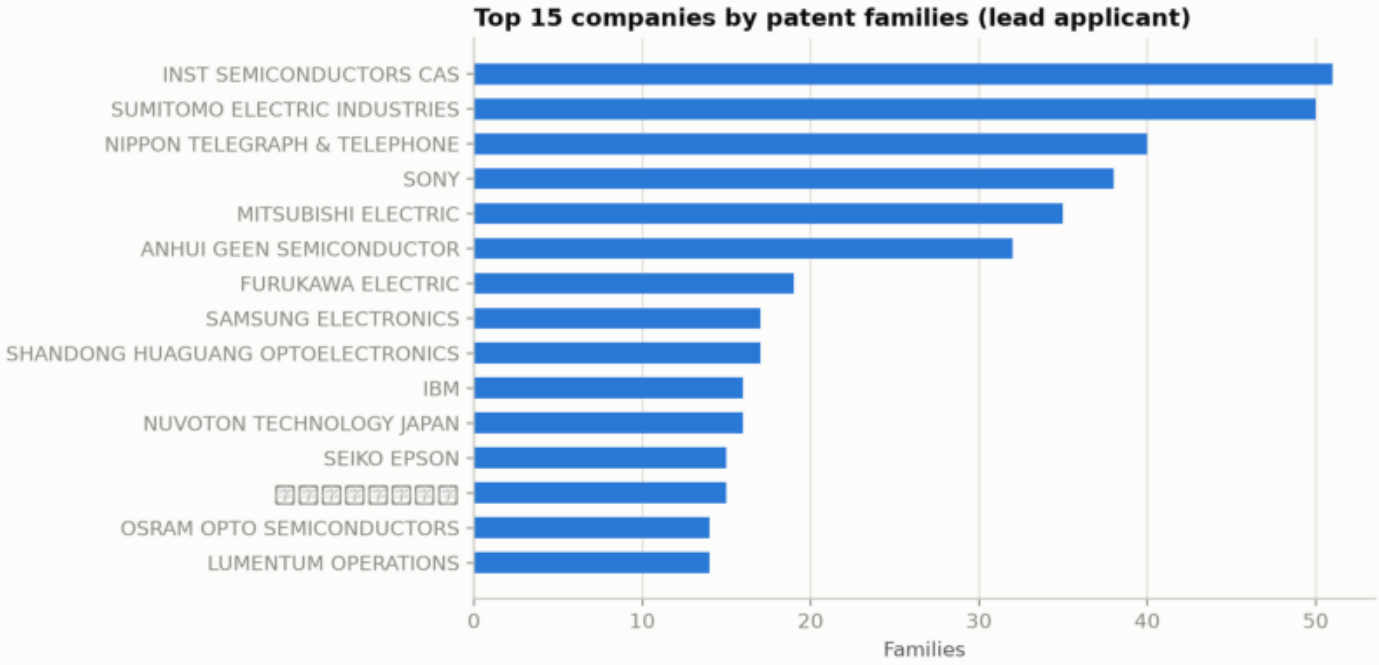
Working set for the charts: 1202 families with priority years 2010 to 2024, without 122 families flagged as GaN or quantum cascade lasers.

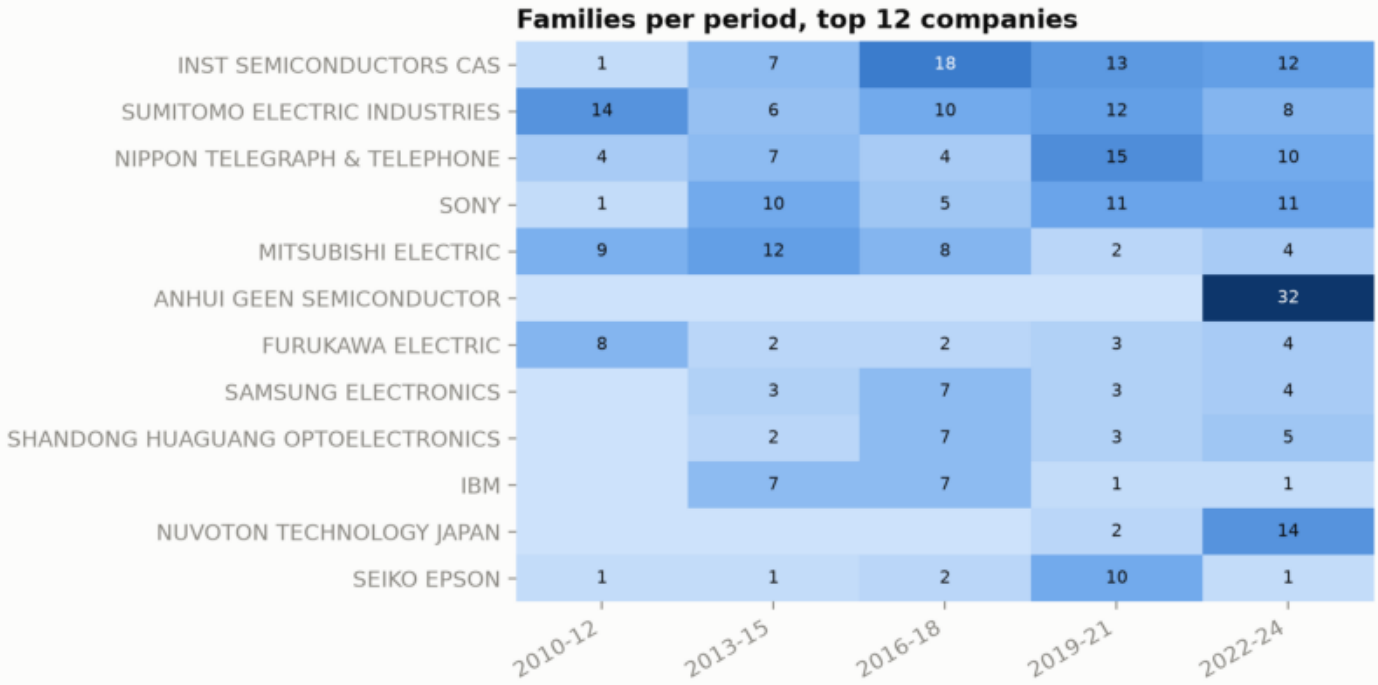
Top companies by families (lead applicant): INST SEMICONDUCTORS CAS (51), SUMITOMO ELECTRIC INDUSTRIES (50), NIPPON TELEGRAPH & TELEPHONE (40), SONY (38), MITSUBISHI ELECTRIC (35).

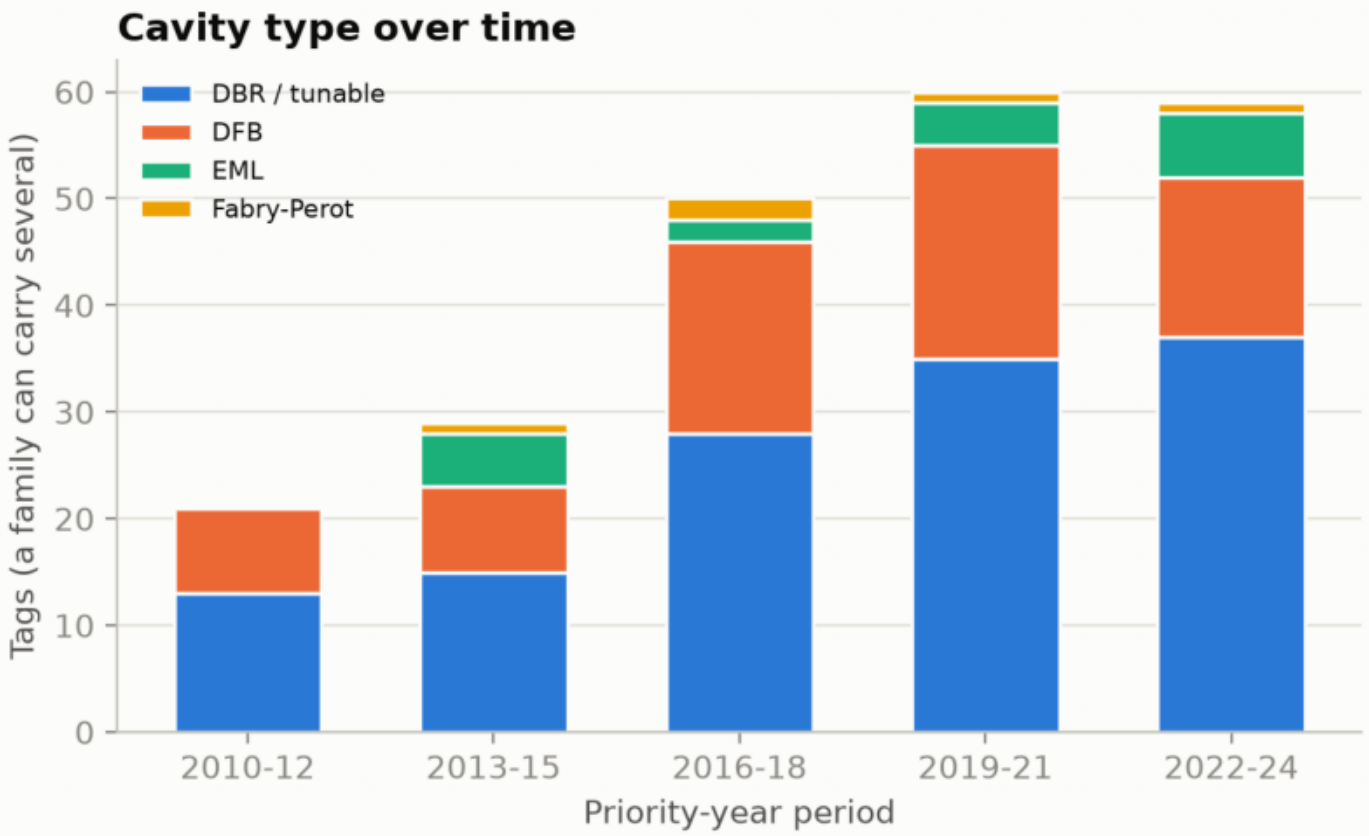
Tag coverage: 48% of families carry a material tag, 27% an emission tag, 5% an application tag.

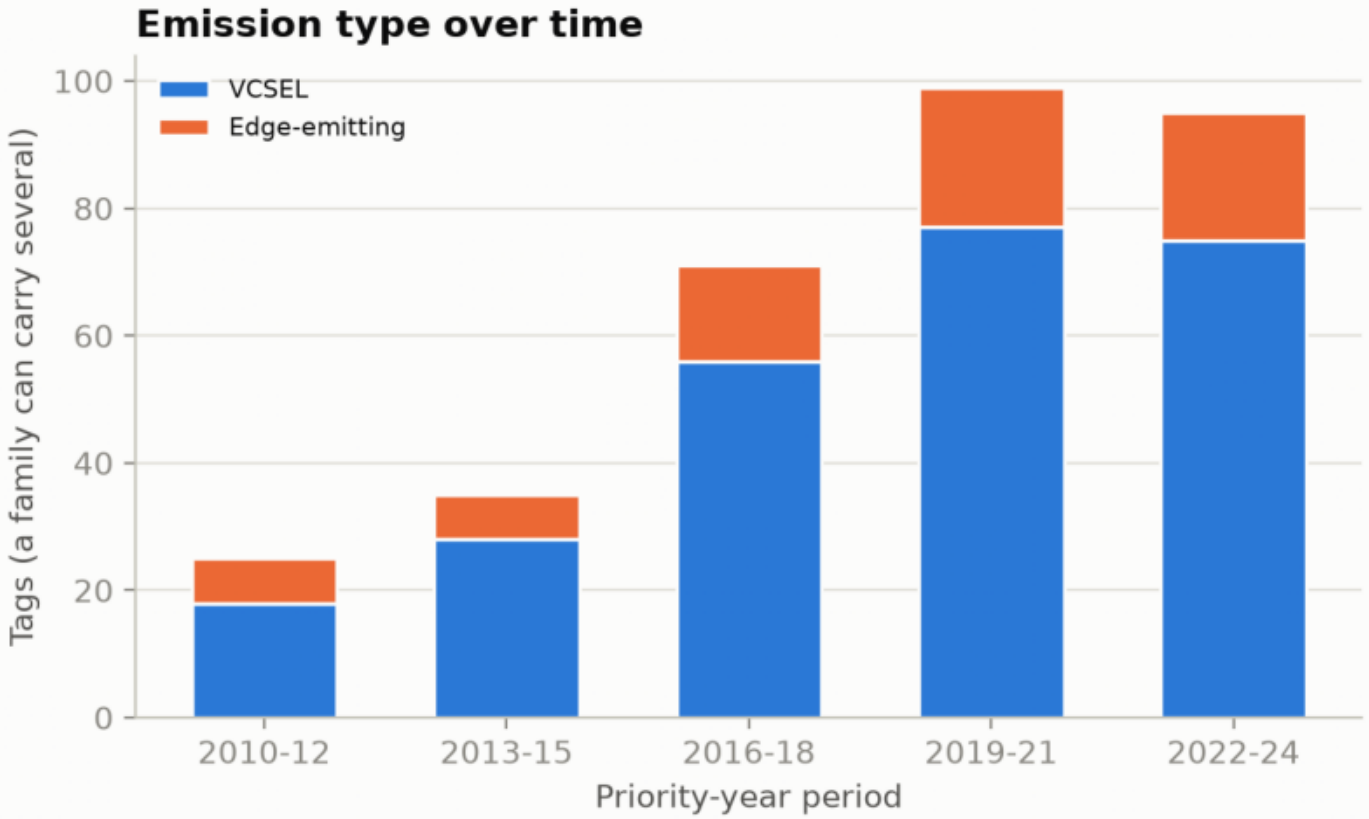
Limits: tags come from keyword rules on titles, abstracts and CPC codes, not from claims. Years before 2010 are not collected. Company names are only lightly cleaned. Counts follow the EPO's one-result-per-application grouping. The latest priority years are incomplete because patents are published about 18 months after filing.

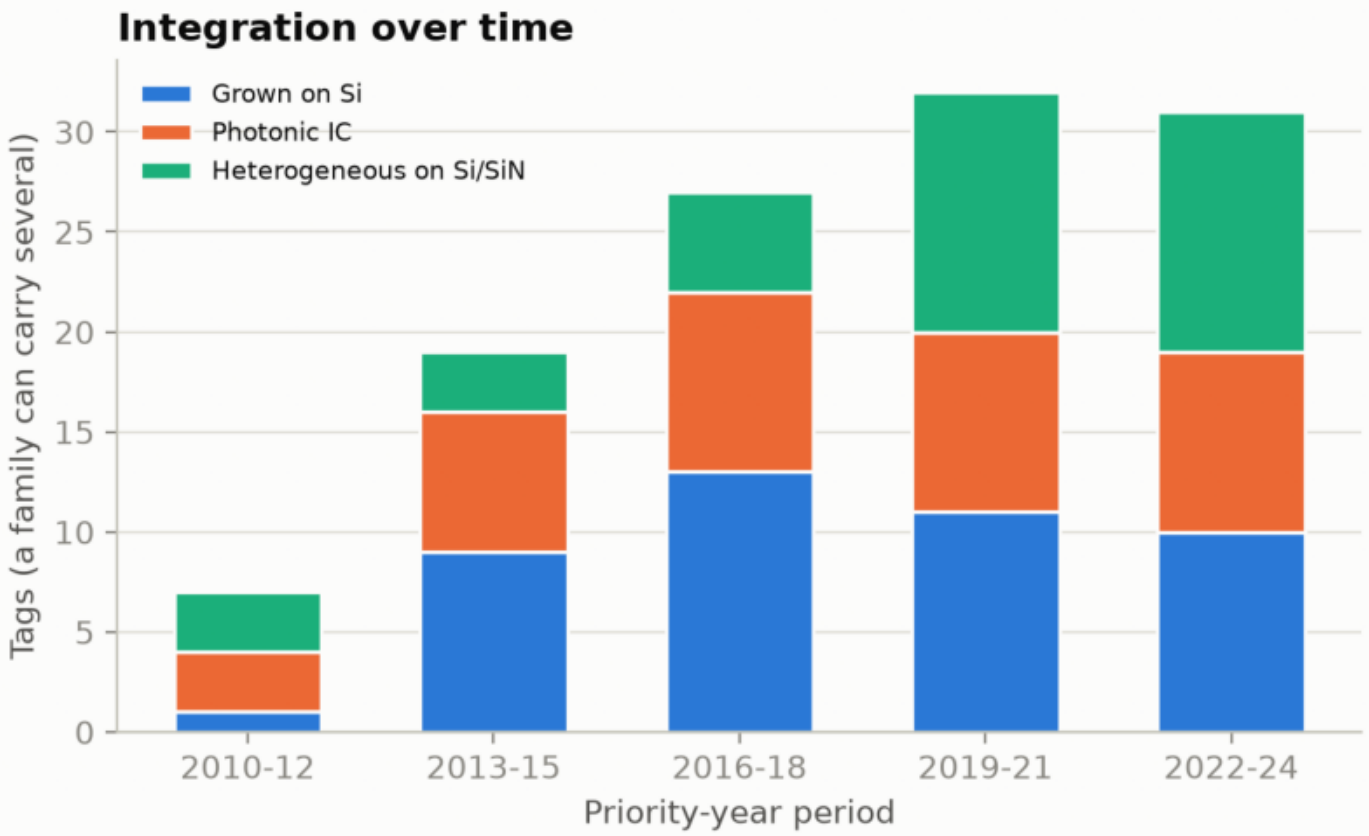

Patent families per priority year
All families
GaAs QD (tagged)
InP (tagged)
Families
120
100
80
60
40
20
0
2010
2012
2014
2016
2018
2020
2022
2024
Priority year

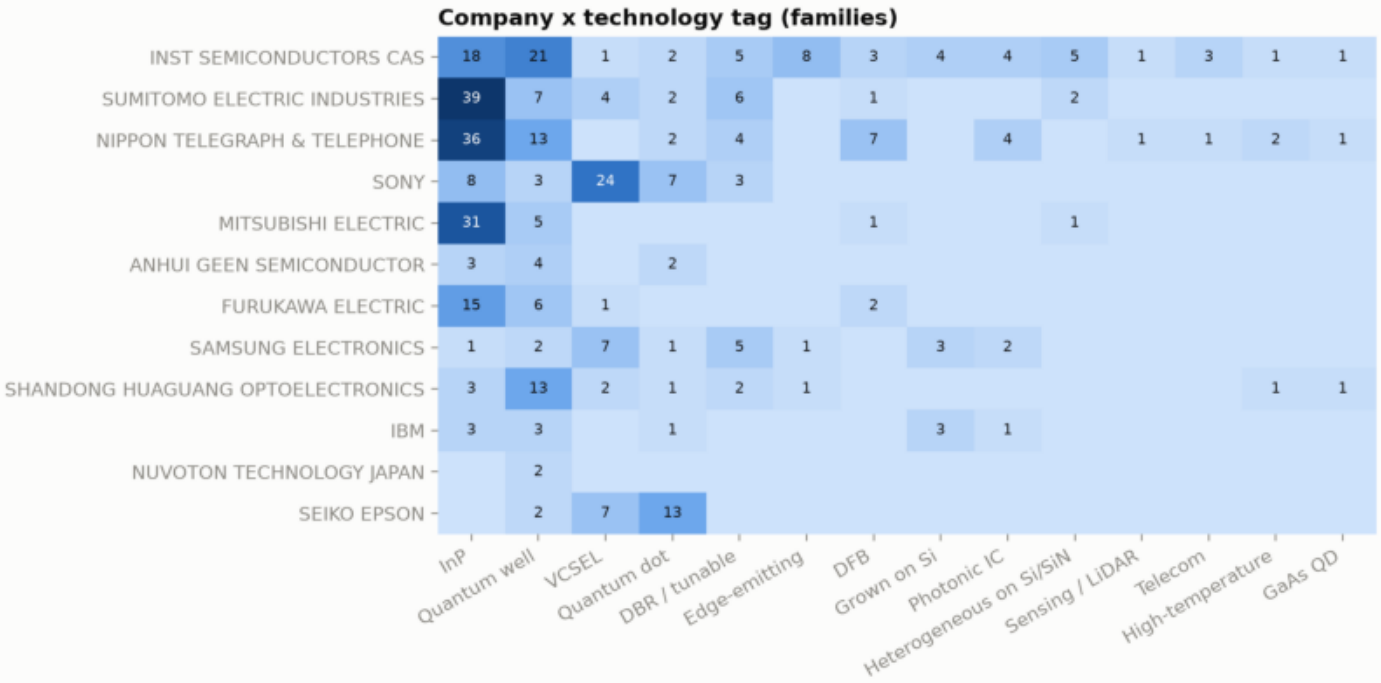

Top 15 companies by patent families (lead applicant)
INST SEMICONDUCTORS CAS
SUMITOMO ELECTRIC INDUSTRIES
NIPPON TELEGRAPH & TELEPHONE
SONY
MITSUBISHI ELECTRIC
ANHUI GEEN SEMICONDUCTOR
FURUKAWA ELECTRIC
SAMSUNG ELECTRONICS
SHANDONG HUAGUANG OPTOELECTRONICS
IBM
NUVOTON TECHNOLOGY JAPAN
SEIKO EPSON
□□□□□□□□
OSRAM OPTO SEMICONDUCTORS
LUMENTUM OPERATIONS
0
10
20
30
40
50
Families

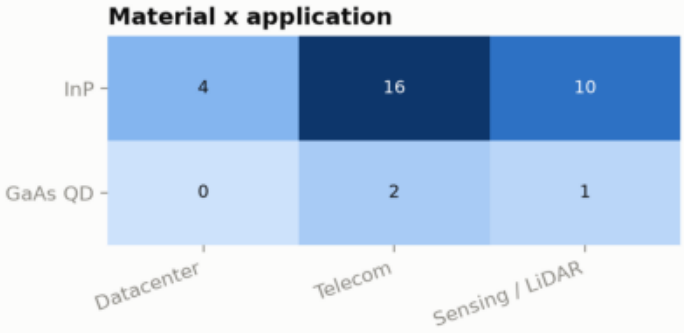

Families per period, top 12 companies
INST SEMICONDUCTORS CAS
SUMITOMO ELECTRIC INDUSTRIES
NIPPON TELEGRAPH & TELEPHONE
SONY
MITSUBISHI ELECTRIC
ANHUI GEEN SEMICONDUCTOR
FURUKAWA ELECTRIC
SAMSUNG ELECTRONICS
SHANDONG HUAGUANG OPTOELECTRONICS
IBM
NUVOTON TECHNOLOGY JAPAN
SEIKO EPSON
1 7 18 13 12
14 6 10 12 8
4 7 4 15 10
1 10 5 11 11
9 12 8 2 4
32
8 2 2 3 4
3 7 3 4
2 7 3 5
7 7 1 1
2 14
1 1 2 10 1
2010-12
2013-15
2016-18
2019-21
2022-24

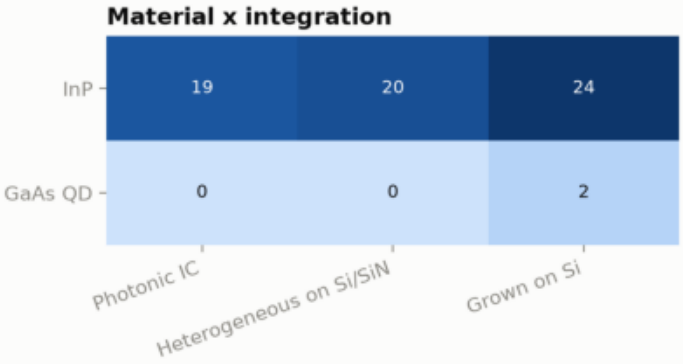

Cavity type over time
DBR / tunable
DFB
EML
Fabry-Perot
Tags (a family can carry several)
0
10
20
30
40
50
60
2010-12
2013-15
2016-18
2019-21
2022-24
Priority-year period

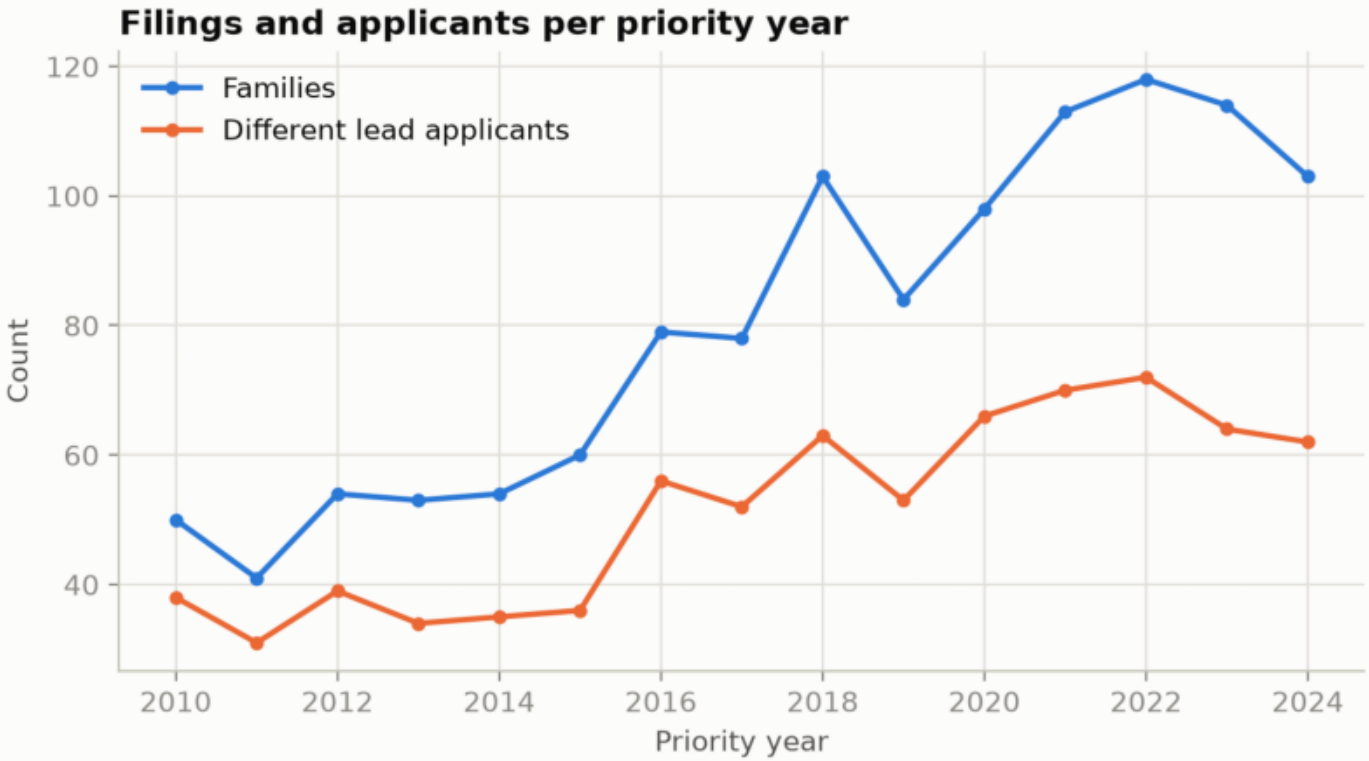

Emission type over time
VCSEL
Edge-emitting
Tags (a family can carry several)
0
20
40
60
80
100
2010-12
2013-15
2016-18
2019-21
2022-24
Priority-year period

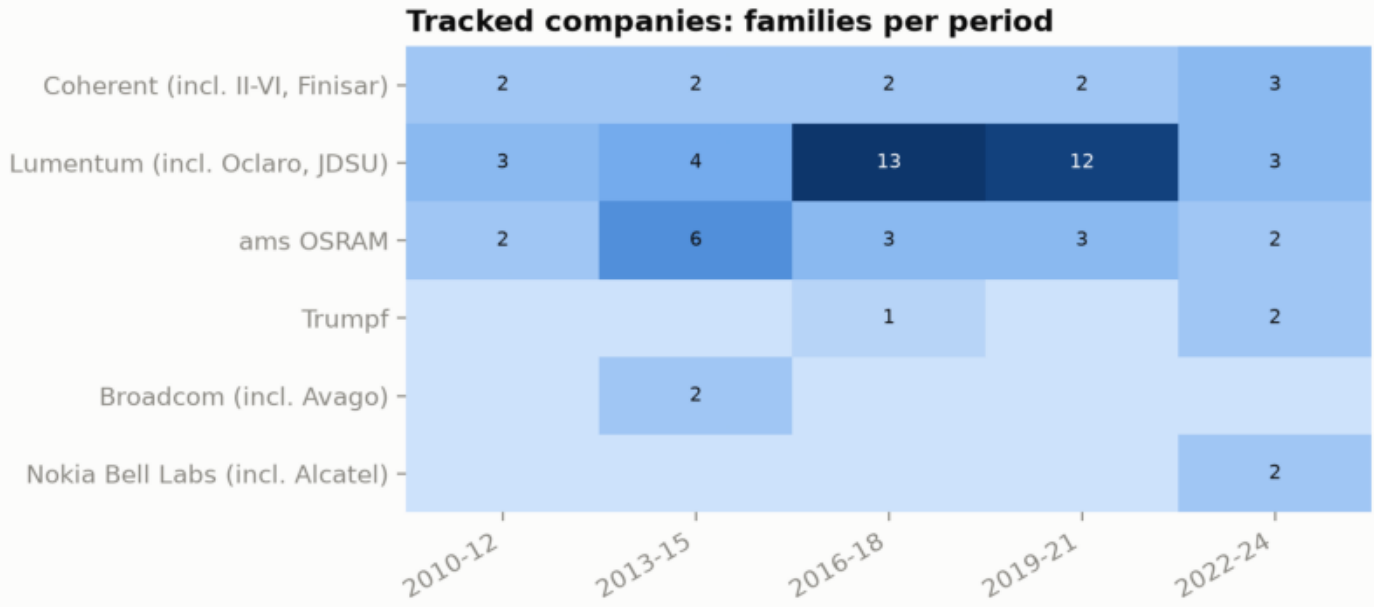

Integration over time
Grown on Si
Photonic IC
Heterogeneous on Si/SiN
Tags (a family can carry several)
0
5
10
15
20
25
30
2010-12
2013-15
2016-18
2019-21
2022-24
Priority-year period

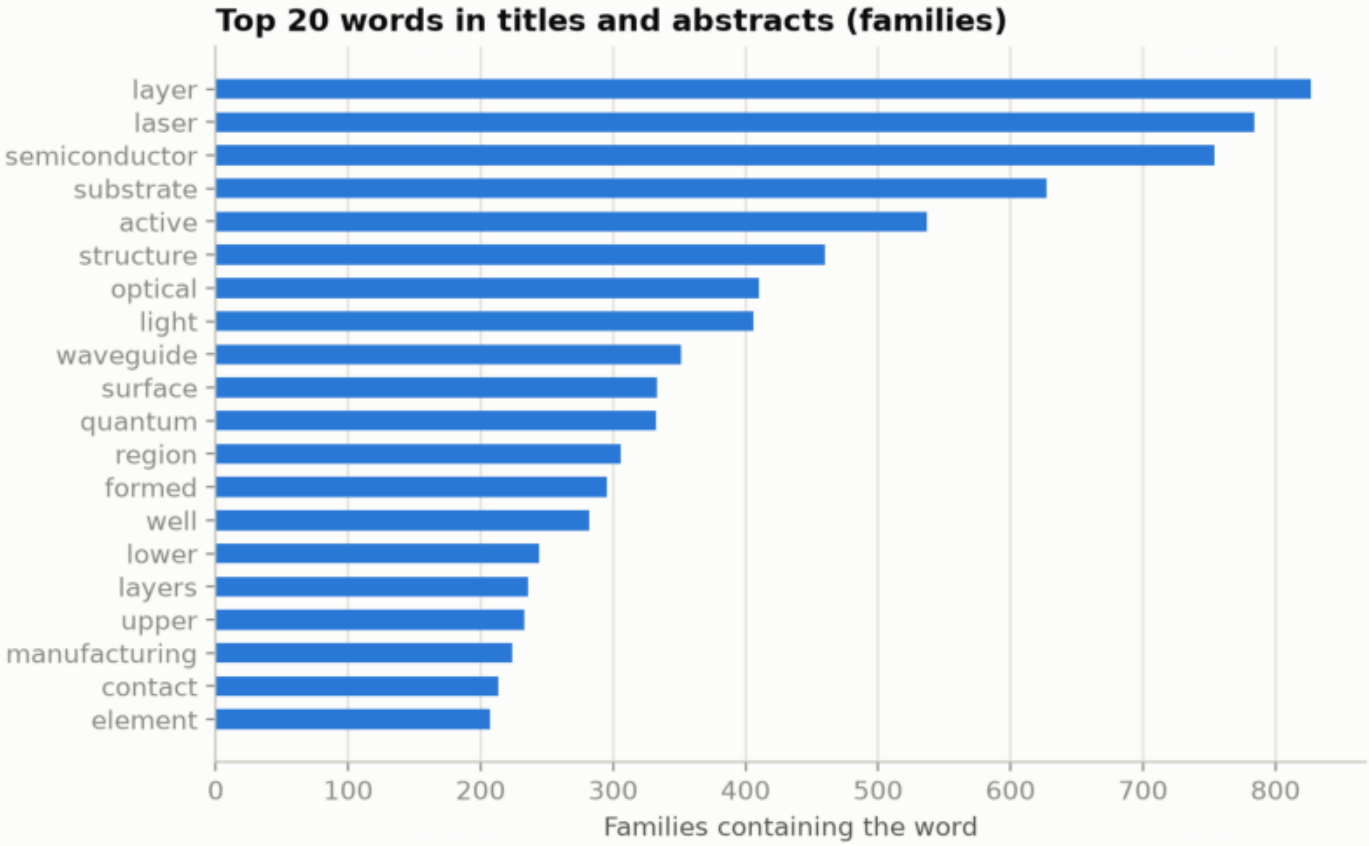

Company x technology tag (families)
INST SEMICONDUCTORS CAS
SUMITOMO ELECTRIC INDUSTRIES
NIPPON TELEGRAPH & TELEPHONE
SONY
MITSUBISHI ELECTRIC
ANHUI GEEN SEMICONDUCTOR
FURUKAWA ELECTRIC
SAMSUNG ELECTRONICS
SHANDONG HUAGUANG OPTOELECTRONICS
IBM
NUVOTON TECHNOLOGY JAPAN
SEIKO EPSON
18 21 1 2 5 8 3 4 4 5 1 3 1 1
39 7 4 2 6 1 2
36 13 2 4 7 4 1 1 2 1
8 3 24 7 3
31 5 1 1
3 4 2
15 6 1 2
1 2 7 1 5 1 3 2
3 13 2 1 2 1 1 1
3 3 1 3 1
2
2 7 13
InP
Quantum well
VCSEL
Quantum dot
DBR / tunable
Edge-emitting
DFB
Grown on Si
Photonic IC
Heterogeneous on Si/SiN
Sensing / LiDAR
Telecom
High-temperature
GaAs QD


Material x application
InP
GaAs QD
4
16
10
0
2
1
Datacenter
Telecom
Sensing / LiDAR


Material x integration
InP
GaAs QD
19
20
24
0
0
2
Photonic IC
Heterogeneous on Si/SiN
Grown on Si


Filings and applicants per priority year
Families
Different lead applicants
Count
120
100
80
60
40
2010
2012
2014
2016
2018
2020
2022
2024
Priority year


Tracked companies: families per period
Coherent (incl. II-VI, Finisar)
Lumentum (incl. Oclaro, JDSU)
ams OSRAM
Trumpf
Broadcom (incl. Avago)
Nokia Bell Labs (incl. Alcatel)
2 2 2 2 3
3 4 13 12 3
2 6 3 3 2
1 2
2
2
2010-12
2013-15
2016-18
2019-21
2022-24


Top 20 words in titles and abstracts (families)
layer
laser
semiconductor
substrate
active
structure
optical
light
waveguide
surface
quantum
region
formed
well
lower
layers
upper
manufacturing
contact
element
0
100
200
300
400
500
600
700
800
Families containing the word

Initial observations

Observation 1: replace this with a finding and its number, for example the growth of families per year.

Observation 2: replace this with a finding about companies.

Observation 3: replace this with a finding about the technology mix.

Activity roughly doubled: families per priority year rose from 55 in 2010 to about 125 a year in 2021 to 2024. 2025 and 2026 are incomplete because patents are published about 18 months after filing.

Japanese and Chinese organisations lead: Sumitomo Electric has the most families (89), then the Institute of Semiconductors CAS (60), Anhui Geen Semiconductor (56), NTT (48) and Mitsubishi Electric (40). Of the six tracked companies, only Lumentum and OSRAM Opto Semiconductors reach the top 15 (17 families each).

Most families carry no technology tag from titles and abstracts (about 80% have no material tag and 95% no application tag), and only 16 families are tagged datacenter. Datacenter and white-space conclusions need claims text and better tagging in Week 2.

The dataset is a broad candidate pool: 2,248 documents form 1,540 families, and 122 families (8%) mention GaN or quantum cascade lasers and are left out of the charts.